

VENANGO, PA, USA

WMO: 725267

Lat: 41.383N

Lon: 79.867W

Elev: 1540

StdP: 13.90

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 94868

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 0.8	(c) 6.6	(d) -7.6	(e) 3.9	(f) 3.3	(g) -1.9	(h) 5.2	(i) 8.4	(j) 23.3	(k) 35.3	(l) 20.6	(m) 32.4	(n) 7.5	(o) 240	(p) 0.511

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.2	85.7	71.8	82.6	69.8	81.3	68.8	74.1	82.1	72.4	80.1	70.8	78.0	7.8	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
72.0	125.1	79.1	70.2	117.6	77.0	68.4	110.4	75.1	38.6	81.7	37.1	79.3	35.8	77.6	80.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
19.5	17.3	15.5	-3.7	89.1	6.3	2.9	-8.3	91.2	-12.0	92.9	-15.5	94.5	-20.1	96.6		
			-4.4	76.9	6.2	1.6	-8.9	78.1	-12.5	79.1	-16.0	80.0	-20.5	81.2		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.3	25.1	27.2	35.4	47.5	57.8	65.5	69.5	68.0	61.7	50.6	39.7	30.6
	DBStd	18.03	12.10	11.53	11.41	9.76	8.46	6.37	5.25	5.05	6.94	8.45	9.19	9.65
	HDD50	3111	774	641	475	160	25	0	0	3	98	329	606	
	HDD65	6539	1236	1058	918	531	254	71	18	28	145	454	759	1067
	CDD50	2498	3	3	23	84	265	465	603	559	354	115	20	4
	CDD65	451	0	0	0	5	30	86	157	121	46	6	0	0
	CDH74	3258	0	0	4	55	278	621	1138	806	324	32	0	0
	CDH80	697	0	0	0	10	43	134	274	170	62	4	0	0
(14) Wind	WSAvg	7.0	8.5	8.1	7.9	8.1	6.9	6.1	5.4	5.0	5.6	6.9	7.8	8.1
(15) Precipitation	PrecAvg	43.2	3.3	2.6	3.3	3.7	3.9	4.0	4.8	4.1	4.1	3.4	3.3	3.4
	PrecMax	55.2	6.7	5.9	6.3	5.9	6.7	6.9	10.1	7.7	8.1	7.6	5.6	6.7
	PrecMin	32.1	1.5	1.4	1.4	1.7	1.4	0.9	1.3	1.6	1.1	1.4	1.4	1.2
	PrecStd	6.3	1.3	1.2	1.4	1.0	1.4	1.7	2.5	1.4	2.0	1.3	1.2	1.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.7	62.8	72.4	80.7	84.0	87.6	89.9	88.2	85.4	78.3	66.4	60.9
		MCWB	53.8	52.6	57.7	61.8	68.6	72.6	74.4	71.8	70.9	67.1	56.4	53.9
	2%	DB	54.4	54.5	64.3	73.3	81.1	83.8	85.7	84.1	81.5	72.7	62.9	54.8
		MCWB	51.4	48.7	52.5	57.8	65.8	70.6	72.8	70.9	68.9	62.1	53.9	51.0
	5%	DB	48.1	48.4	59.3	69.7	77.2	81.2	82.4	81.5	78.5	69.6	58.7	50.0
		MCWB	45.0	43.4	50.2	56.1	63.5	68.8	70.4	69.3	67.2	60.9	51.7	45.9
	10%	DB	42.7	43.5	54.1	63.9	72.9	77.7	81.0	79.1	73.4	64.2	54.7	45.0
		MCWB	38.7	38.0	46.8	52.1	61.3	66.4	69.4	67.6	64.5	57.4	48.9	41.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.6	55.1	59.0	64.4	71.3	74.3	77.2	75.5	73.5	69.5	59.4	56.5
		MCDB	59.4	60.0	69.1	75.6	80.1	82.7	85.7	83.6	81.6	75.9	64.4	59.4
	2%	WB	51.6	50.0	55.3	60.4	68.4	72.3	74.7	73.3	70.8	65.6	55.9	51.5
		MCDB	54.4	53.9	62.2	69.7	77.8	81.4	82.7	80.6	78.4	70.6	60.3	54.1
	5%	WB	45.5	44.2	51.6	57.5	66.1	70.5	72.8	71.4	68.7	61.9	53.0	46.3
		MCDB	47.6	48.1	58.3	66.1	74.3	78.7	80.2	77.8	75.0	66.8	57.6	48.4
	10%	WB	38.4	38.8	46.4	54.4	63.5	68.4	70.9	69.8	66.7	58.7	49.6	41.3
		MCDB	41.8	43.2	52.5	62.2	70.6	75.3	77.7	75.9	72.1	64.1	55.4	45.3
(35) Mean Daily Temperature Range	5% DB	MDBR	13.4	15.3	17.1	20.5	20.4	19.1	19.2	18.9	18.9	17.3	14.2	11.7
		MCDBR	17.8	21.4	24.6	27.0	24.4	21.9	21.1	21.7	22.9	23.0	20.7	17.7
		MCWBR	15.7	17.0	15.6	15.9	12.8	11.7	10.5	10.8	11.5	13.6	14.7	15.3
		MCDBR	17.4	20.4	22.0	22.4	20.8	18.9	18.1	17.6	18.2	19.1	17.7	16.5
	5% WB	MCWBR	16.5	17.6	15.4	15.2	12.3	11.1	9.9	10.2	10.2	13.1	14.5	15.8
(40) Clear-Sky Solar Irradiance	taub		0.293	0.323	0.340	0.368	0.404	0.436	0.434	0.425	0.374	0.350	0.310	0.293
	taud		2.438	2.334	2.380	2.349	2.257	2.198	2.221	2.237	2.361	2.433	2.523	2.495
	Ebn at Noon		274	280	288	286	277	267	266	265	273	267	266	266
	Edh at Noon		24	32	34	38	43	45	44	42	35	29	23	22
(44) All-Sky Solar Radiation	RadAvg		443	708	1019	1395	1640	1793	1819	1628	1311	843	551	348
	RadStd		40	91	111	149	154	156	140	93	135	86	77	35

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (41 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+194	N/A

Nomenclature: See separate page